

O'NEILL, NE, USA

WMO: 725566

Lat: 42.470N

Lon: 98.688W

Elev: 619

StdP: 94.11

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04957

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-18.2	-25.3	0.4	-20.7	-22.5	0.5	-17.8	16.3	-7.3	14.8	-5.6	4.6	320	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	34.8	21.6	32.6	21.6	31.1	21.0	24.1	30.2	23.3	29.3	22.5	28.5	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.5	27.4	21.4	17.4	26.6	20.6	16.5	26.1	76.3	30.3	72.7	29.6	69.4	27.9	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.8	10.5	DB	-25.4	37.6	2.5	2.4	-27.2	39.3	-28.7	40.7	-30.1	42.1	-31.9	43.9	(4)
(5)			WB	-25.7	25.7	2.4	0.9	-27.4	26.4	-28.8	26.9	-30.1	27.4	-31.9	28.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-4.5	-2.9	2.8	8.8	14.7	20.5	23.9	22.3	17.9	10.0	3.0	-3.5	(6)	
	DBStd	11.43	7.20	6.98	7.02	5.51	4.84	4.01	3.40	3.24	4.82	5.41	6.13	6.56	(7)	
	HDD10.0	1854	449	362	239	85	13	1	0	0	3	66	220	417	(8)	
	HDD18.3	3702	708	594	482	289	134	24	3	6	67	261	459	676	(9)	
	CDD10.0	1666	0	1	16	49	157	314	432	382	239	67	10	0	(10)	
	CDD18.3	473	0	0	1	3	21	87	176	129	52	4	0	0	(11)	
	CDH23.3	4902	0	0	13	54	246	869	1868	1209	569	68	5	0	(12)	
	CDH26.7	1950	0	0	1	12	69	306	842	489	213	18	1	0	(13)	
(14) Wind	WSAvg	4.7	5.0	5.0	5.2	5.5	5.0	4.4	4.0	3.9	4.5	4.9	4.8	4.8	(14)	
(15) Precipitation	PrecAvg	641	14	16	37	73	100	98	66	72	57	61	26	17	(15)	
	PrecMax	945	39	37	79	153	202	225	126	157	124	152	61	72	(16)	
	PrecMin	477	2	5	2	22	6	37	18	11	15	1	0	2	(17)	
	PrecStd	130	10	9	22	36	49	48	30	45	32	43	20	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	18.6	25.2	28.0	31.9	34.7	38.8	37.2	34.6	28.9	23.2	14.8	(19)	
		MCWB	6.7	8.7	12.5	15.1	18.6	22.0	21.2	21.3	19.3	16.1	11.3	6.8	(20)	
	2%	DB	10.6	13.0	21.1	24.2	28.4	32.2	35.5	33.3	31.9	25.0	18.8	11.2	(21)	
		MCWB	4.8	6.2	11.1	13.4	18.1	21.7	22.2	21.5	19.9	14.6	10.0	5.7	(22)	
	5%	DB	7.3	9.7	17.1	21.3	26.1	30.1	33.0	31.4	29.0	22.2	15.3	7.5	(23)	
		MCWB	2.9	4.4	10.0	12.4	16.9	20.7	22.4	21.2	19.1	13.6	8.4	3.0	(24)	
	10%	DB	4.8	6.5	13.1	18.2	23.6	28.0	31.3	29.3	27.0	19.0	12.2	5.0	(25)	
		MCWB	1.7	2.8	7.5	10.7	15.9	19.6	21.9	20.9	18.0	12.2	6.9	1.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	9.7	15.2	16.8	21.1	24.4	25.7	24.8	23.0	18.4	13.2	8.9	(27)	
		MCDB	14.8	17.2	19.5	24.4	28.1	30.6	31.8	30.6	29.1	25.5	19.0	13.7	(28)	
	2%	WB	5.0	6.9	12.5	14.8	19.4	23.0	24.6	23.6	21.7	16.3	10.8	5.3	(29)	
		MCDB	10.1	12.4	18.6	21.8	26.0	29.5	30.8	29.3	27.6	21.6	17.4	10.7	(30)	
	5%	WB	3.2	4.8	10.2	13.1	17.9	21.9	23.8	22.8	20.6	14.7	8.9	3.5	(31)	
		MCDB	6.8	9.4	17.0	20.0	24.2	28.3	29.9	28.2	26.6	19.9	15.2	7.2	(32)	
	10%	WB	1.8	2.9	7.8	11.5	16.6	20.8	22.9	21.8	19.4	12.8	6.8	1.8	(33)	
		MCDB	4.5	6.2	13.3	17.4	22.5	26.9	28.9	27.4	25.2	18.6	11.9	4.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	10.7	11.8	12.0	11.5	11.6	12.3	12.1	13.1	12.4	11.8	9.9	(35)	
		MCDBR	14.1	15.7	18.1	17.5	15.1	13.8	14.6	15.0	16.0	17.1	17.2	13.8	(36)	
	5% WB	MCWBR	9.2	9.6	9.8	8.8	7.3	6.1	5.4	5.9	6.6	8.6	9.7	9.0	(37)	
		MCWBR	13.6	14.8	16.8	15.1	13.1	12.1	12.4	12.0	13.2	14.1	16.6	13.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.265	0.278	0.304	0.342	0.365	0.379	0.392	0.394	0.353	0.321	0.277	0.265	(40)	
		taud	2.493	2.482	2.456	2.359	2.381	2.379	2.368	2.344	2.420	2.486	2.529	2.485	(41)	
	Ebn at Noon		897	937	945	926	908	893	876	863	879	871	874	862	(42)	
		Edh at Noon	71	85	99	117	118	119	119	118	102	84	68	65	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.01	2.94	3.90	4.98	5.88	6.68	6.79	5.75	4.65	3.23	2.24	1.69	(44)	
	RadStd		0.18	0.20	0.35	0.29	0.46	0.45	0.27	0.40	0.33	0.36	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.66	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page